JOINT ADMISSION TEST FOR MASTERS • IIT JAM MATHEMATICS (MA)

IIT JAM 2011 Official Paper

Category: Classic Era (2005–2014) • Official Mock Test Series

TOTAL QUESTIONS	TOTAL MARKS	TEST DURATION	PAPER PATTERN
15	90 M	60 Min	MCQ

Section / Type	Questions	Marking Scheme	Negative Marking	Format / Details
Multiple Choice (MCQ)	15	+1 / +2 Marks	−1/3 / −2/3 Marks	4 Options (1 Correct)

IMPORTANT INSTRUCTIONS FOR CANDIDATES

1. **Authentic Examination PYQs:** This mock paper contains verified official IIT JAM Mathematics questions and high-resolution problem statements.
2. **Section A (MCQ):** Each question has four choices (A), (B), (C), (D), out of which **only one** is correct. Wrong answers carry negative penalty (1/3 of positive marks).
3. **Section B (MSQ):** Each question has four choices (A), (B), (C), (D), out of which **one or more than one** choice(s) may be correct. Full marks are awarded only if all correct choices and zero incorrect choices are chosen. No partial marks and no negative marks.
4. **Section C (NAT):** Numerical answers are real numbers entered via virtual keyboard. Full credit is awarded if the numerical answer falls within the official accepted interval. No negative marking.
5. **Master Answer Key:** Complete official answers and acceptance bounds are provided at the end of this document.
6. **Online Interactive Simulator:** Practice this test with virtual calculator, timers, and instant scoring analytics at **vaibhavgeometer.github.io**.

IIT JAM Mathematics Mock Test Series • Curated by Vaibhav Geometer

Question 1 (JAM 2011 • Q1 • ID: JAM_2011_Q1) MCQ +6 M Neg: −2.0

Q.1 Let $a_n = \sum_{k=1}^{n} \frac{n}{n^2+k}$, for $n \in \mathbb{N}$. Then the sequence $\{a_n\}$ is

(A) Convergent
(B) Bounded but not convergent
(C) Diverges to ∞
(D) Neither bounded nor diverges to ∞

Question 2 (JAM 2011 • Q2 • ID: JAM_2011_Q2) MCQ +6 M Neg: −2.0

Q.2 The number of real roots of the equation $x^3 + x - 1 = 0$ is

(A) 0 (B) 1 (C) 2 (D) 3

Question 3 (JAM 2011 • Q3 • ID: JAM_2011_Q3) MCQ +6 M Neg: −2.0

Q.3 The value of $\lim_{n\to\infty} \sum_{k=1}^{n} \frac{1}{\sqrt{n^2+kn}}$ is

(A) $2(\sqrt{2}-1)$ (B) $2\sqrt{2}-1$ (C) $2-\sqrt{2}$ (D) $\frac{1}{2}(\sqrt{2}-1)$

Question 4 (JAM 2011 • Q4 • ID: JAM_2011_Q4) MCQ +6 M Neg: −2.0

Q.4 Let V be the region bounded by the planes $x=0,\ x=2,\ y=0,\ z=0$ and $y+z=1$. Then the value of the integral $\iiint_V y\,dx\,dy\,dz$ is

(A) $\frac{1}{2}$ (B) $\frac{4}{3}$ (C) 1 (D) $\frac{1}{3}$

Question 5 (JAM 2011 • Q5 • ID: JAM_2011_Q5) MCQ +6 M Neg: −2.0

Q.5 The solution $y(x)$ of the differential equation $\frac{d^2y}{dx^2}+4\frac{dy}{dx}+4y=0$ satisfying the conditions $y(0)=4,\ \frac{dy}{dx}(0)=8$ is

(A) $4e^{2x}$ (B) $(16x+4)e^{-2x}$ (C) $4e^{-2x}+16x$ (D) $4e^{-2x}+16xe^{2x}$

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Question 6 (JAM 2011 • Q6 • ID: JAM_2011_Q6) MCQ +6 M Neg: −2.0

Q.6 If y^a is an integrating factor of the differential equation $2xy\,dx-(3x^2-y^2)\,dy=0$, then the value of a is

(A) -4 (B) 4 (C) -1 (D) 1

Question 7 (JAM 2011 • Q7 • ID: JAM_2011_Q7) MCQ +6 M Neg: −2.0

Q.7 Let $\vec{F}=ay\hat{i}+z\hat{j}+x\hat{k}$ and C be the positively oriented closed curve given by $x^2+y^2=1,\ z=0$. If $\oint_C \vec{F}\cdot d\vec{r}=\pi$, then the value of a is

(A) -1 (B) 0 (C) $\frac{1}{2}$ (D) 1

Question 8 (JAM 2011 • Q8 • ID: JAM_2011_Q8) MCQ +6 M Neg: −2.0

Q.8 Consider the vector field $\vec{F}=(ax+y+a)\hat{i}+\hat{j}-(x+y)\hat{k}$, where a is a constant. If $\vec{F}\cdot\text{curl}\,\vec{F}=0$, then the value of a is

(A) -1 (B) 0 (C) 1 (D) $\frac{3}{2}$

Question 9 (JAM 2011 • Q9 • ID: JAM_2011_Q9) **MCQ** +6 M Neg: −2.0

Q.9 Let G denote the group of all 2×2 invertible matrices with entries from $\mathbb{R}$. Let

$$H_1 = \{A \in G : \det(A) = 1\} \text{ and } H_2 = \{A \in G : A \text{ is upper triangular}\}.$$

Consider the following statements:

P : H_1 is a normal subgroup of G
Q : H_2 is a normal subgroup of G.

Then

(A) Both P and Q are true
(B) P is true and Q is false
(C) P is false and Q is true
(D) Both P and Q are false

Question 10 (JAM 2011 • Q10 • ID: JAM_2011_Q10) **MCQ** +6 M Neg: −2.0

Q.10 For $n \in \mathbb{N}$, let $n\mathbb{Z} = \{nk : k \in \mathbb{Z}\}$. Then the number of units of $\mathbb{Z}/11\mathbb{Z}$ and $\mathbb{Z}/12\mathbb{Z}$, respectively, are

(A) 11, 12
(B) 10, 11
(C) 10, 4
(D) 10, 8

Question 11 (JAM 2011 • Q11 • ID: JAM_2011_Q11) **MCQ** +6 M Neg: −2.0

Q.11 Let A be a 3×3 matrix with trace $(A) = 3$ and $\det(A) = 2$. If 1 is an eigenvalue of A, then the eigenvalues of the matrix $A^2 - 2I$ are

(A) $1,\ 2(i-1),\ -2(i+1)$
(B) $-1,\ 2(i-1),\ 2(i+1)$
(C) $1,\ 2(i+1),\ -2(i+1)$
(D) $-1,\ 2(i-1),\ -2(i+1)$

Question 12 (JAM 2011 • Q12 • ID: JAM_2011_Q12) MCQ +6 M Neg: −2.0

Q.12 Let $T:\mathbb{R}^n \to \mathbb{R}^n$ be a linear transformation, where $n \geq 2$. For $k \leq n$, let

$$E = \{v_1, v_2, \ldots, v_k\} \subseteq \mathbb{R}^n \text{ and } F = \{Tv_1, Tv_2, \ldots, Tv_k\}.$$

Then

(A) If E is linearly independent, then F is linearly independent
(B) If F is linearly independent, then E is linearly independent
(C) If E is linearly independent, then F is linearly dependent
(D) If F is linearly independent, then E is linearly dependent

Question 13 (JAM 2011 • Q13 • ID: JAM_2011_Q13) MCQ +6 M Neg: −2.0

Q.13 For $n \neq m$, let $T_1:\mathbb{R}^n \to \mathbb{R}^m$ and $T_2:\mathbb{R}^m \to \mathbb{R}^n$ be linear transformations such that T_1T_2 is bijective. Then

(A) rank $(T_1) = n$ and rank $(T_2) = m$
(B) rank $(T_1) = m$ and rank $(T_2) = n$
(C) rank $(T_1) = n$ and rank $(T_2) = n$
(D) rank $(T_1) = m$ and rank $(T_2) = m$

Question 14 (JAM 2011 • Q14 • ID: JAM_2011_Q14) MCQ +6 M Neg: −2.0

Q.14 The set of all x at which the power series $\sum_{n=1}^{\infty} \frac{n}{(2n+1)^2}(x-2)^{3n}$ converges is

(A) $[-1, 1)$
(B) $[-1, 1]$
(C) $[1, 3)$
(D) $[1, 3]$

Question 15 (JAM 2011 • Q15 • ID: JAM_2011_Q15) MCQ +6 M Neg: −2.0

Q.15 Consider the following subsets of $\mathbb{R}$:

$$E = \left\{ \frac{n}{n+1} : n \in \mathbb{N} \right\}, \qquad F = \left\{ \frac{1}{1-x} : 0 \le x < 1 \right\}.$$

Then

(A) Both E and F are closed

(B) E is closed and F is NOT closed

(C) E is NOT closed and F is closed

(D) Neither E nor F is closed

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OFFICIAL MASTER ANSWER KEY & EVALUATION RUBRIC

IIT JAM 2011 Official Paper • All Official Answers & Acceptance Ranges

Q. No.	Type	Marks	Official Key
1	MCQ	+6	**A**
2	MCQ	+6	**B**
3	MCQ	+6	**A**
4	MCQ	+6	**D**
5	MCQ	+6	**B**
6	MCQ	+6	**A**
7	MCQ	+6	**A**
8	MCQ	+6	**D**
9	MCQ	+6	**B**
10	MCQ	+6	**C**
11	MCQ	+6	**D**
12	MCQ	+6	**B**
13	MCQ	+6	**D**
14	MCQ	+6	**C**
15	MCQ	+6	**C**

SCORING METHODOLOGY & EVALUATION GUIDELINES

- **Multiple Choice Questions (MCQ):** Full marks for correct single choice. Negative marks deducted for each incorrect attempt (-1/3 for 1-mark questions, $-2/3$ for 2-mark questions). Unattempted questions award zero.
- **Multiple Select Questions (MSQ):** Full marks if and only if all correct choices are marked and no incorrect choice is selected. Zero marks for partial selection or any incorrect choice. No negative marking.
- **Numerical Answer Type (NAT):** Any numerical value falling strictly within the specified official range (e.g., [a, b]) receives full credit. No negative marking.
- **Marks to All (MTA):** Questions officially designated as MTA award full marks to all candidates regardless of attempt.